Build guide 1: tools and steel

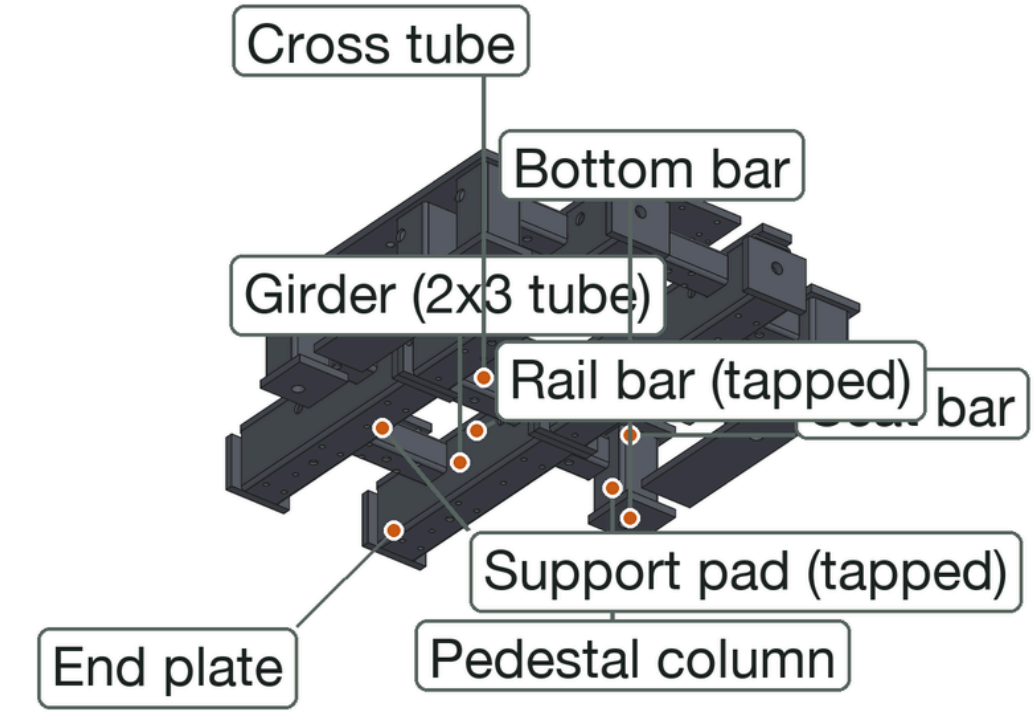
DIY CNC epoxy-granite mill · October 2026

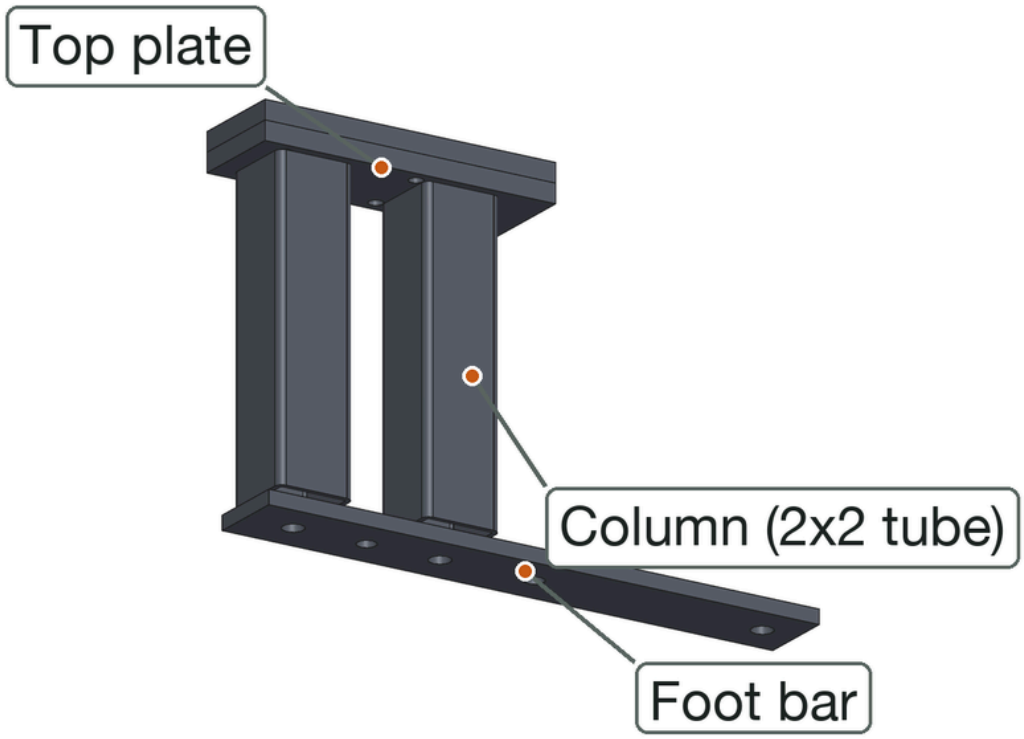
0. Tools, space and safety

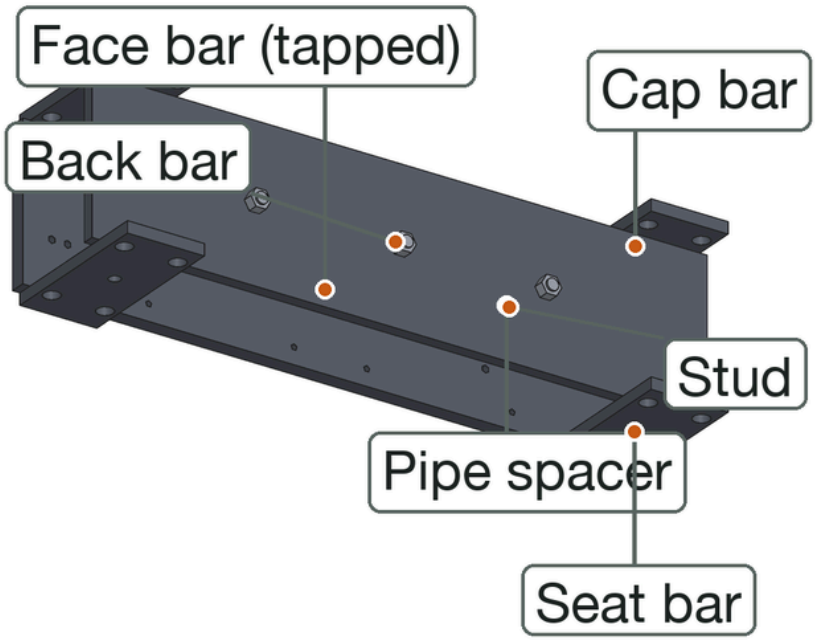
- **Space:** a level bench or floor area of about 1.2 x 1.5 m at 20-25 °C for a week. Epoxy cures slowly and badly below 18 °C.
- **Tools:** drill press, chop saw or hacksaw, angle grinder with flap disc, taps (M4, M5, M6), drill/mixer with a paddle (or a small cement mixer), kitchen scale + a 50 kg scale, vibration source (a table vibrator, a concrete poker, or a reciprocating sander with the pad removed pressed against the mold), straightedge (at least 600 mm, ideally ground), feeler gauges, dial indicator with a magnetic base, precision level, torque wrench.
- **Safety:** nitrile gloves and long sleeves for epoxy (skin sensitisation is permanent); an N95 dust mask when pouring the whiting; a P100 respirator when grinding or sanding cured EG (the quartz sand and gravel in it make silica dust; work wet if you can); eye protection when cutting steel. Mix in a ventilated space.
- **3D printing (PETG):** the mold kit (brackets, saddles, holders, bushings), the core posts, formers and fillers, the parts that get cast in (column rings, tube plugs, drain liner) and the machine's printed parts. All of it is in `print/`, split to fit a Bambu A1 mini, and listed in the print guide: 1.8 kg of PETG (sliced), inside two 1 kg rolls, and about 120 hours of printing; 1.1 kg and 84 hours if the base's bevel strips are a ripped 2x3 instead. The base pocket cores are cut from one sheet of 2 in XPS foam board instead. Print one bracket and one saddle first and check them on your board; print a column ring early and test-fit it in your actual tube.
- **Foam cutting:** a fine-tooth saw or a hot-wire cutter, a long knife, a sanding block, packing tape and double-sided carpet tape.

1. Buy and prepare the steel

The frame's backbone is steel: a skeleton in each casting, bolted together from **stock flat bar and tube** and clamped with threaded rod, then cast into epoxy granite, which damps it and stiffens it further. Every joint between the castings, and everything that bolts on (rails, ballscrew supports), is steel to steel. Nothing is welded, nothing is custom-cut, and nothing is threaded into the epoxy granite.


Cross tube
Bottom bar
Girder (2x3 tube)
Rail bar (tapped)
bar
Support pad (tapped)
End plate
Pedestal column

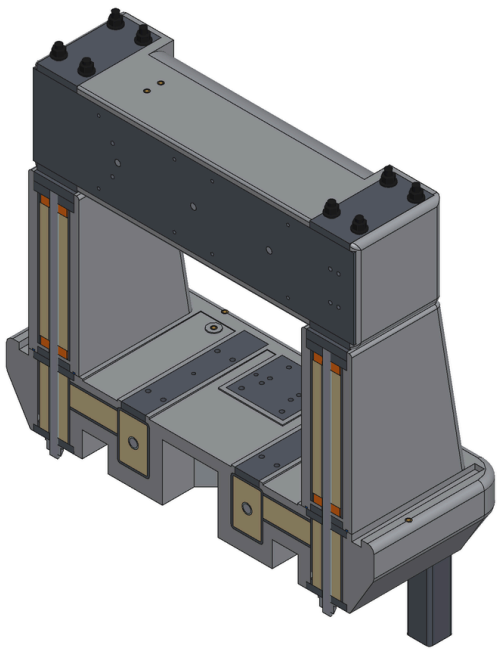

Top plate
Column (2x2 tube)
Foot bar



1. **Buy the steel cut to length.** `docs/1-sourcing/steel/README.md` has the cut list by stock size and every part (about 58 pieces, 62 kg). It's all common sizes: 3/8 in flat bar in 1-1/2, 2, 2-1/2 and 5 in widths, 1/2 x 2-1/2 and 1/2 x 5 in flat bar, 2 x 3 x 1/8 in rectangular tube, and 2 x 2 x 1/8 and 1-1/2 x 1-1/2 x 1/8 in square tube. A local steel yard or Metal Supermarkets will cut it for you (cheapest); online sellers sell it in fixed lengths you cut with a chop saw or hacksaw. Plus, bought at length: 2 x 2 in tube, 4 x 8-1/8 in (the upright columns), and 1-1/2 x 1-1/2 in tube, 4 x 6 in (legs). A saw cut within 1/8 in is fine everywhere (the report's "Steel cut to length" table lists where each error goes). **Have the two 5/8 x 6 in plates (the X carriage and the Z plate) and the two Z risers Blanchard ground** on both faces, flat and parallel, before you drill them: the rail blocks sit on them metal to metal, with no epoxy bed to take up mill scale or bow. Metal Supermarkets grinds what it sells at some stores; otherwise any grinding shop. Nothing else needs it: the Y and X risers are bedded on epoxy.
2. **Drill and tap from the templates.** Print `docs/1-sourcing/steel/steel-templates.pdf` at actual size (check the 100 mm square), tape each part's outline to the part, center-punch every crosshair and drill on a drill press. The legend on each page gives the drill, and which holes to tap: 1/2-13 in the upright top plates' lower bars (the column tie rods screw in) and 3/8-16 for the beam studs; M5 in the base rail bars and the beam face bar; M6 in the support pads and the beam face bar. Mark one of each pair and use it as a drilling jig for the other.
3. **Bolt each skeleton together on the bench:**
   - **Upright** (two): just the parts for now: foot bar, two columns (a printed ring in each end) and the top plate's two bars. The column tie rods clamp it at assembly.
   - **Beam:** screw the three 113 mm studs into the face bar, a pipe nipple over each, the back bar over them and a nut on each; the seat and cap bars go in the mold on the cap-screw rods.
   - **Base** (one grid per half): each girder (2 x 3 tube) between its end plates, its rail bar bolted on top with 1/4-20 x 4 in bolts (nuts under the tube) and the draw rod through it; the pedestals (seat bar, the 2 x 2 tube spacers, bottom bar) held together with the tie rods for now; the cross tubes between the girders and from each girder to its pedestal, each clamped by its 1/2-13 rod and two nuts; the support pads bolted onto their cross tubes. Glue the four 3/8 x 1 in dowel pins into the rear half's split end plates, half their length out.

- **Sleeves:** before the draw rods and the pedestals' tie rods go in, slide a vinyl sleeve (1/2 in ID x 5/8 in OD) over each rod, cut to the inside length of the tube it runs in (the girder between its end plates, the pedestal column between its bars). These rods come out later; the sleeve keeps the sand out of their way. The cross rods stay in for good and need none.
- **Sand:** turn the grid upside down, as it goes in the mold, and fill every base tube with dry pool filter sand through its 3/8 in fill hole (the girders' and cross tubes' bottom face, the pedestal columns' outer face by the bottom bar) with a small funnel, tapping the tube with a mallet until no more goes in. Tape over each hole (packing tape) and wipe off the sand outside: epoxy won't bond to it.

4. **Rail risers** (cold-rolled 1018, 1-1/4 in wide, ordered cut to the rail lengths): Y 3/4 in x 600 mm (2), X 5/8 in x 400 mm (2), Z 5/8 in x 300 mm (2, ground with the plates). The rails are used at full length.
   - Clamp each rail on its riser, centered, and use the rail's own holes as the drill template; drill and tap M4 (60 mm pitch). Mark which rail goes on which riser.
   - Y and X risers also get the hold-down holes, midway between rail holes and under the rail: 5.5 mm through with a 9.5 mm x 4.5 mm deep counterbore from the top. Positions from the riser center: Y at 0, ±120 and ±240 mm, X at ±30 and ±150 mm. They go into the tapped rail bars (base) and face bar (beam).
   - Stone or lightly sand off burrs; keep the top faces flat.
5. **Clean all the steel:** wire-brush or flap-disc to dull gray metal, then degrease with acetone just before casting. If it will sit more than a day or two, give it a thin coat of epoxy primer (not on the faces the mold screws or the rails bear on). Paint only the part of each leg that will hang below the base (after casting).
6. **Studs and hardware:** nothing to cut. The 1/2-13 rods are grade B7 studs bought at length: per upright 2 x 14-1/4 in (in the columns) and 1 x 6 in (at the toe); 2 x 27 in for the base draw rods; 4 x 10-3/4 and 6 x 8-1/2 in for the base cross rods. The 3/8-16 ones: 8 beam studs of 7-1/2 in and 3 ties of 4-1/2 in. One end is set flush and the far nut's tail takes the rest (at least 4.6 mm of it). If a length had to be cut from rod, deburr the end. Have 19 x M5 cast-in inserts ready: 11 for the enclosure, 4 for the way-cover rollers and 2 for the Y cable chain in the base, 2 for the cable-chain mount in the beam. They hold only those add-ons; nothing structural threads into the epoxy granite. Cast them all in, even if the enclosure and covers come later.



The cast-in steel for each base half (castings drawn see-through):

